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# ============================================================
# IPL / CRICKET DATA ANALYTICS PROJECT
# Using:
# Python + Pandas + NumPy + Matplotlib
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# ============================================================
# STEP 1 : CREATE SAMPLE DATASET
# ============================================================

data = {
    "Player": [
        "Virat", "Rohit", "Dhoni", "Gill", "Surya",
        "Virat", "Rohit", "Dhoni", "Gill", "Surya"
    ],

    "Team": [
        "RCB", "MI", "CSK", "GT", "MI",
        "RCB", "MI", "CSK", "GT", "MI"
    ],

    "Runs": [
        82, 45, 30, 95, 60,
        70, 100, 40, 88, 72
    ],
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    "Balls": [
        50, 30, 25, 55, 35,
        45, 60, 20, 50, 40
    ],

    "Wickets": [
        0, 0, 1, 0, 0,
        0, 0, 2, 0, 0
    ],

    "Season": [
        2024, 2024, 2024, 2024, 2024,
        2025, 2025, 2025, 2025, 2025
    ],

    "Result": [
        "Win", "Lose", "Win", "Win", "Lose",
        "Win", "Win", "Lose", "Win", "Win"
    ]
}

# CREATE DATAFRAME
df = pd.DataFrame(data)

# ============================================================
# STEP 2 : DISPLAY DATA
# ============================================================

print("\n========== IPL DATA ==========\n")
print(df)
```

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# ============================================================
# STEP 3 : STRIKE RATE CALCULATION
# ============================================================

df["Strike_Rate"] = (df["Runs"] / df["Balls"]) * 100

print("\n========== STRIKE RATE ==========\n")
print(df[["Player", "Runs", "Balls", "Strike_Rate"]])

# ============================================================
# STEP 4 : HIGHEST RUN SCORERS
# ============================================================

print("\n========== TOP RUN SCORERS ==========\n")

top_runs = df.groupby("Player")["Runs"].sum().sort_values(ascending=False)
print(top_runs)

# ============================================================
# STEP 5 : BEST STRIKE RATE
# ============================================================

print("\n========== BEST STRIKE RATE ==========\n")

best_sr = df.groupby("Player")["Strike_Rate"].mean().sort_values(ascending=False)
print(best_sr)
```

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# ============================================================
# STEP 6 : TEAM PERFORMANCE
# ============================================================

print("\n========== TEAM PERFORMANCE ==========\n")

team_runs = df.groupby("Team")["Runs"].sum()
print(team_runs)

# ============================================================
# STEP 7 : MATCH WIN ANALYSIS
# ============================================================

print("\n========== MATCH RESULTS ==========\n")

wins = df[df["Result"] == "Win"]["Team"].value_counts()
print(wins)

# ============================================================
# STEP 8 : PLAYER RANKING
# ============================================================

print("\n========== PLAYER RANKING ==========\n")

ranking = df.groupby("Player")["Runs"].sum().sort_values(ascending=False)
print(ranking)
```

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# ============================================================
# STEP 9 : PLAYER PERFORMANCE GRAPH
# ============================================================

plt.figure(figsize=(8,5))
top_runs.plot(kind='bar')
plt.title("Player Total Runs")
plt.xlabel("Player")
plt.ylabel("Runs")
plt.savefig("player_runs_chart.png")
plt.show()

# ============================================================
# STEP 10 : TEAM WIN ANALYSIS
# ============================================================

plt.figure(figsize=(8,5))
wins.plot(kind='bar')
plt.title("Team Wins")
plt.xlabel("Team")
plt.ylabel("Wins")
plt.savefig("team_wins_chart.png")

plt.show()



# ============================================================
# STEP 11 : STRIKE RATE GRAPH
# ============================================================
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plt.figure(figsize=(8,5))

best_sr.plot(kind='line', marker='o')
plt.title("Player Strike Rate")
plt.xlabel("Player")
plt.ylabel("Strike Rate")
plt.savefig("strike_rate_chart.png")
plt.show()

# ============================================================
# STEP 12 : SEASON COMPARISON
# ============================================================

print("\n========== SEASON COMPARISON ==========\n")

season_runs = df.groupby("Season")["Runs"].sum()
print(season_runs)
plt.figure(figsize=(8,5))
season_runs.plot(kind='pie', autopct='%1.1f%%')
plt.title("Season Runs Comparison")
plt.ylabel("")
plt.savefig("season_comparison_chart.png")
plt.show()

# ============================================================
# STEP 13 : SAVE REPORT
# ============================================================

df.to_csv("ipl_analysis_report.csv", index=False)
print("\nIPL report saved as ipl_analysis_report.csv")
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# ============================================================
# STEP 14 : FINAL SUMMARY
# ============================================================

print("\n========== FINAL SUMMARY ==========\n")

print("Project Name : IPL Cricket Data Analytics")
print("Technology : Python + Pandas + NumPy + Matplotlib")
print("Total Players :", df["Player"].nunique())
print("Total Teams :", df["Team"].nunique())

print("\nProject Completed Successfully")
```